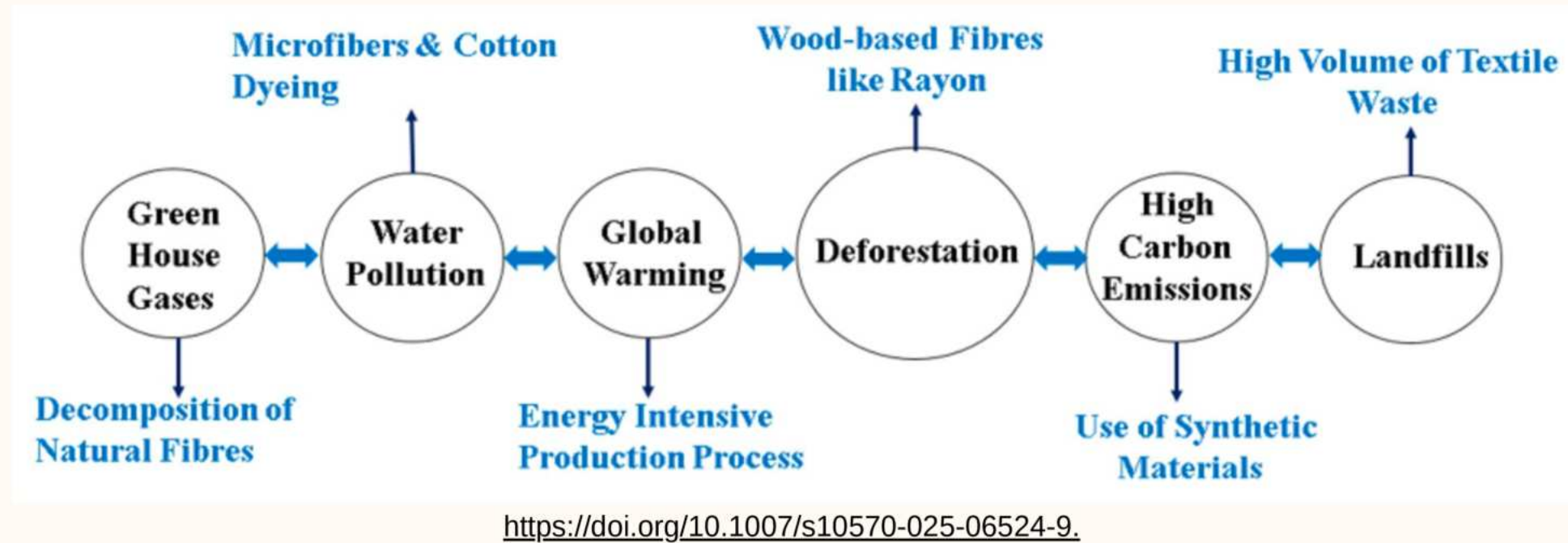


Background



- **92 million tons** of textile waste are produced, accumulating in landfills.
- **Fast fashion** worsens the problem through mass production to follow trends and reduction in usage duration.

<https://oig.org/california/articles/the-fashion-industry-waste-is-dramatically-contributing-to-climate-change/>



<https://doi.org/10.1007/s10570-025-06524-9>

- Combustion of fossil fuels (oil, gas, & coal) are responsible for **90% of CO₂**.
- They represent **80% of total energy supply** globally.



<https://vcresearch.berkeley.edu/news/evidence-supporting-epas-obligation-regulate-greenhouse-gases-stronger-ever>

To address both issues textile waste can be repurposed as a feedstock for bioethanol production.

Current Research Gap:

- Existing research mainly focuses on **pretreatment** methods that break down cellulose crystallinity in pure cotton.
 - Pretreatment is important because it can increase yields and improve enzyme efficiency.
- The focus on just pure cotton textiles is less applicable because modern textiles increasingly contain **synthetic fibers**.
- This study investigates address this gap by determining the feasibility of different cotton blends as feedstocks.

Objectives

Purpose: To determine which cotton blend is most affected by the pretreatment process.

Variables: The independent variable is the type of cotton blend (rayon, nylon, polyester). The dependent variable is the glucose and bioethanol concentrations.

Control: The control is utilizing 100% cotton fabric as a feedstock.

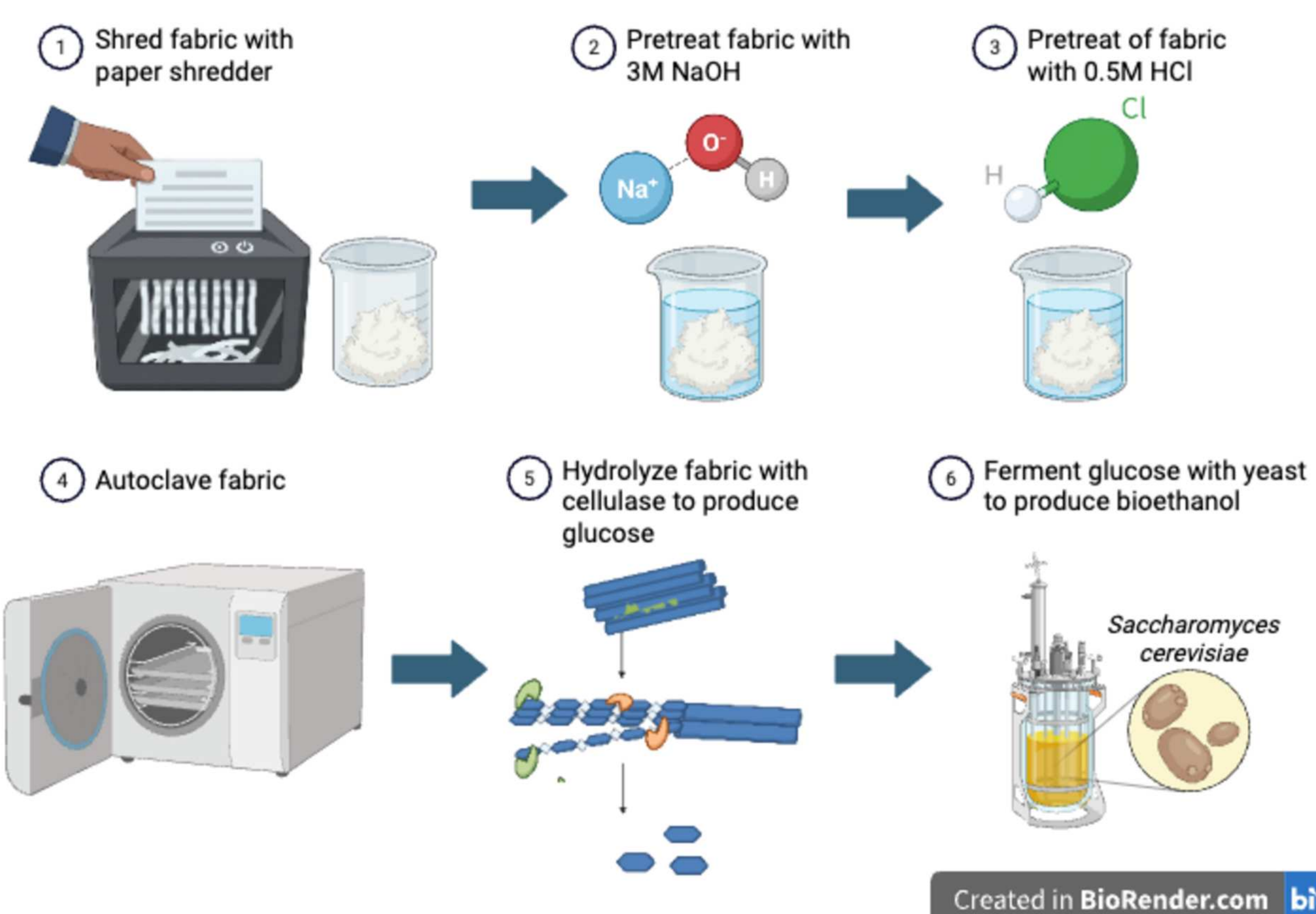
Hypothesis: Cotton-rayon will be most affected by pretreatment due to its structural similarity to cotton, allowing for more effective processing. Cotton-nylon is expected to be moderately affected because nylon's strong amide linkages increase resistance. Cotton-polyester is predicted to be the least affected, as its PET structure contains strong ester bonds and aromatic rings that resist chemical treatment.

Constants: enzymes (cellulase and yeast), duration and concentration of NaOH and HCl for pretreatment

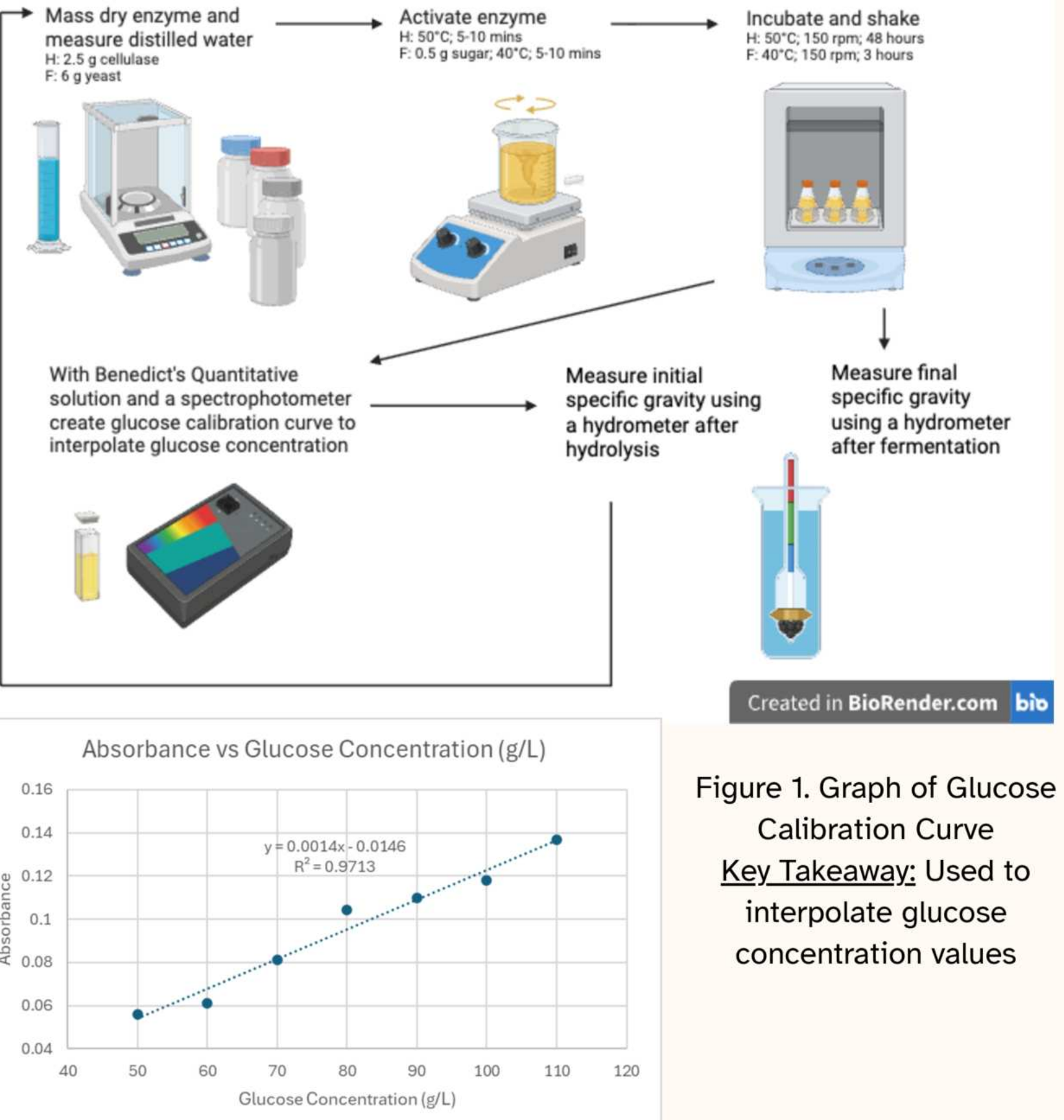
Fashion to Fuel: Evaluating Different Cotton Blends as Viable Feedstocks in Bioethanol Production

Methodology

Procedure Overview



Detailed Hydrolysis and Fermentation Procedure



Results

Pair	P-value
All	0.09
Pure - Rayon	0.54
Pure - Nylon	0.85
Pure - Polyester	0.08
Rayon - Nylon	1.00
Rayon - Polyester	1.00
Nylon - Polyester	1.00

Table 1. Kruskal-Wallis Test Results Comparing Glucose Concentrations

Pair	P-value
All	0.09
Pure - Rayon	0.33
Pure - Nylon	0.10
Pure - Polyester	0.42
Rayon - Nylon	1.00
Rayon - Polyester	1.00
Nylon - Polyester	1.00

Table 2. Kruskal-Wallis Test Results Comparing Bioethanol Vapor Concentrations

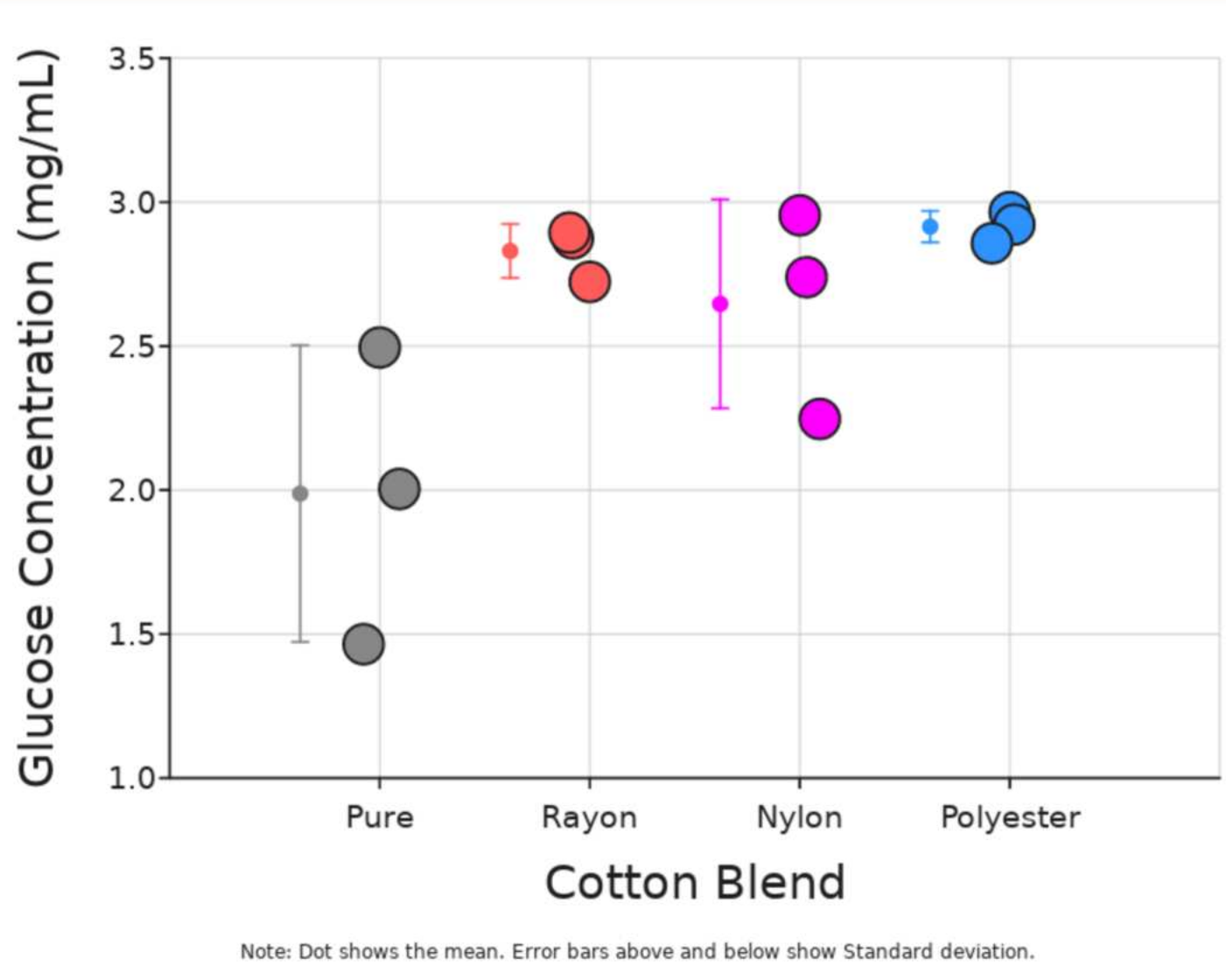


Figure 2. Glucose Concentration vs Cotton Blend

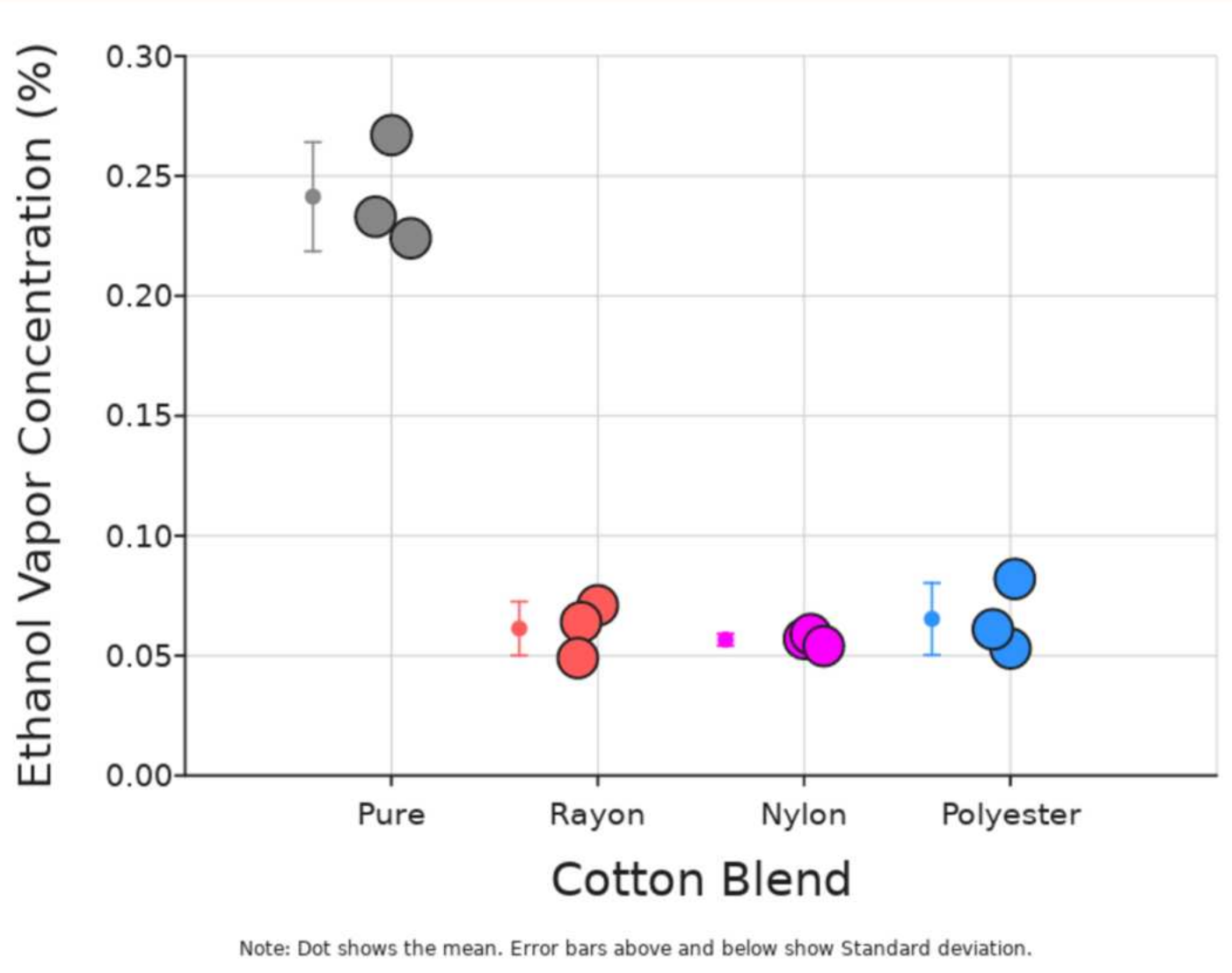


Figure 3. Bioethanol Vapor Concentration vs Cotton Blend

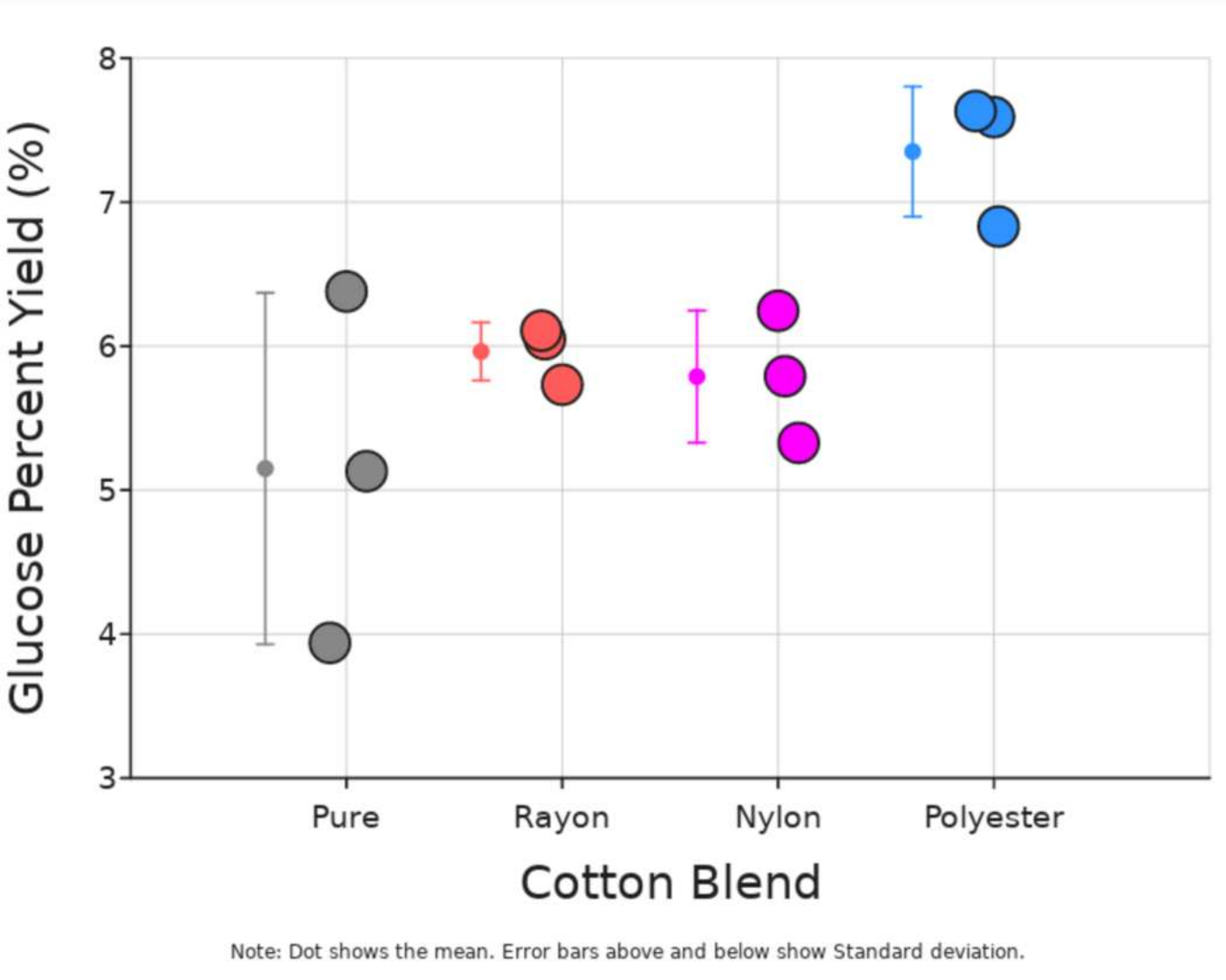


Figure 4. Glucose Percent Yield vs Cotton Blend

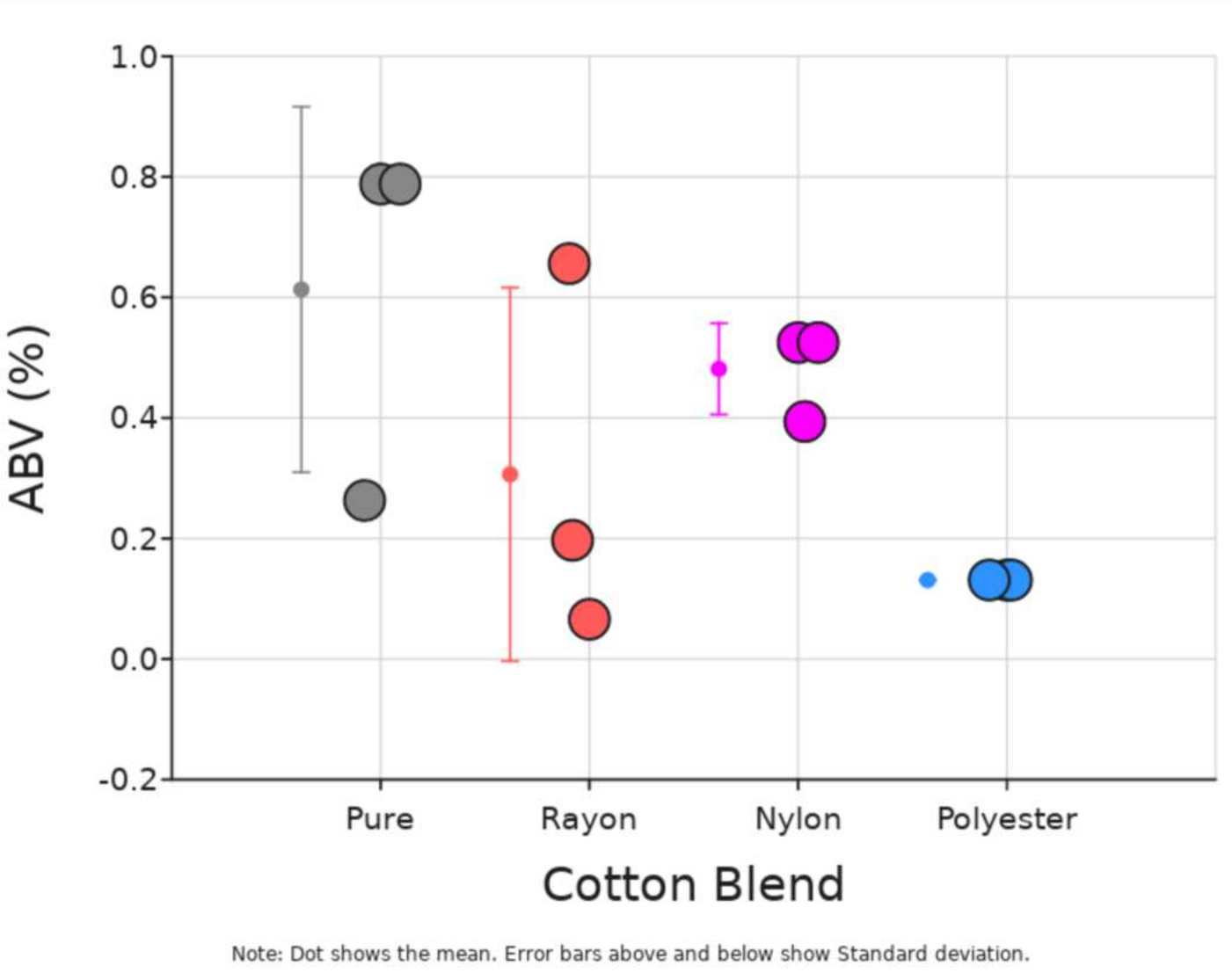


Figure 4. Alcohol by Volume vs Cotton Blend

Discussion

Glucose

- The Kruskal-Wallis test indicated no statistically significant differences in glucose concentration among the experimental groups ($p = 0.09$).
- Post hoc analysis showed:
 - The greatest difference occurred between pure cotton and polyester blends ($p = 0.08$).
 - Comparisons between cotton and nylon ($p = 0.85$) and cotton and rayon ($p = 0.54$) were not significant.
 - No significant differences were observed among the cotton blend groups ($p = 1.00$).
- Overall, **no statistically significant differences** were found; however, pure cotton generally exhibited the lowest glucose concentration, which was not consistent with the original hypothesis.

Bioethanol

- The Kruskal-Wallis test showed no statistically significant differences in bioethanol vapor concentration among the groups ($p = 0.09$).
- Post hoc analysis revealed:
 - The largest difference was between cotton and nylon blends ($p = 0.10$).
 - Comparisons between cotton and polyester ($p = 0.42$) and cotton and rayon ($p = 0.33$) were not significant.
 - No significant differences were found among the cotton blend groups ($p = 1.00$).
- Overall, **no statistically significant differences** in bioethanol production were observed across the experimental groups; however, pure cotton generally had the highest concentrations.

Reflection

- These preliminary findings suggest that **cotton blends may perform comparably to pure cotton** as feedstocks for bioethanol production.
 - Potential implications for improving the economic feasibility and scalability by **broadening the range of usable textile materials**.
- Additional trials and larger sample sizes are necessary to strengthen the reliability of the current results.

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